

Counting cells

A hemocytometer is a specialized microscope slide with a precision-etched grid to determine the number of cells in a known volume. The most common variant for mammalian cells is the Neubauer or “Improved Neubauer” chamber. By staining with a viability dye such as Trypan blue, the proportion of live and dead cells can be determined simultaneously.

Risk assessment

- Work with human-derived material or transgenic cell lines (BSL-1 or BSL-2)
- ▷ Wear gloves, safety glasses, lab coat
- Collect and dispose waste after inactivation as REGULATED MEDICAL WASTE



Reviewed: Mar 14, 2023

Procedures

>> Cell counting with a Neubauer cell chamber

- Neubauer cell chamber
- 0.4% Trypan blue solution, 10 μL (R)
- Hand tally

- (1.) *Optional:* Gently mix the cell suspension with an equal volume of 0.4% Trypan blue. 

This is why: Trypan blue is a diazo dye excluded from live cells, allowing simultaneous viability assessment.

Hint: Automated cell counters use the same Trypan blue exclusion principle and accept 10 μL samples in disposable slides. They reduce user-to-user variability and are preferred for high-throughput work. Follow the manufacturer's instructions for sample preparation and calibration.

- (2.) Clean the hemocytometer and coverslip with a lint-free tissue to remove dust and debris.
- (3.) Moisten the raised glass rails of the hemocytometer. Place the coverslip over the counting surface.
- Note:* The coverslip is thicker than standard types to withstand surface tension and ensure correct chamber depth.
- (4.) Carefully load about 10 μL of the sample into the chamber. The void will fill by capillary action.
- (5.) Place the hemocytometer on the microscope stage and focus. Cells should be evenly distributed and not overlapping.  
- Using the hand tally, count the cells in the four corner grids of the chamber.
 - Exclude cells on the left or bottom border, count cells on the right and top border (or vice versa).
 - Tally unstained (live) and stained (dead) cells separately.

Quality assurance: Repeat with a more concentrated sample if the overall tally is 200 cells or less. 

Hint: Most hemocytometers have nine large 100 μm^2 grids that are arranged in a 3 $\times$ 3 pattern. The four corner grids with 16 small squares are used to count white blood cells and mammalian cells; the center grid is used for smaller particles such as red blood cells, platelets, or cell nuclei.

- (6.) Average the number of cells across the counted grids and multiply by $1 \times 10^4 \text{ mL}^{-1}$ to determine the cell concentration. Be sure to account for any dilution factor introduced by Trypan blue.

Hint: For example, if the average count across four corner grids is 150 cells and the sample was diluted 1:2 with Trypan blue, the concentration is $3.0 \times 10^6 / \text{mL}$ cells.

This is why: The Neubauer and “Improved Neubauer” chambers have a depth of 100 μm between the counting grid and coverslip. Other chamber designs may differ.

Troubleshooting

Cell counting with a Neubauer cell chamber

In Step 5:

- Cells are unevenly distributed under the coverslip
 - Ensure chamber and coverslip are clean and properly assembled.
 - Mix cell suspension gently but thoroughly before loading.

Recipes

Trypan blue solution, 0.4%

Amount	Ingredient	Stock	Final
0.4 g	Trypan Blue [72-57-1]	960.8 g/mol	0.4%
10 mL	Phosphate-buffered saline (PBS), pH 7.4	10 ×	
To 100 mL	Water, reagent-grade		

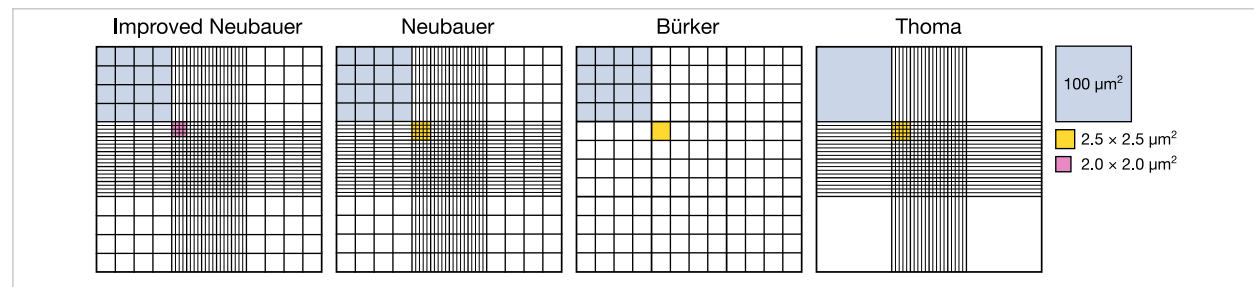
Dissolve the dye in 80% of the final volume; bring to a slow boil, cool down and make up with water.

0.4% Trypan blue solution

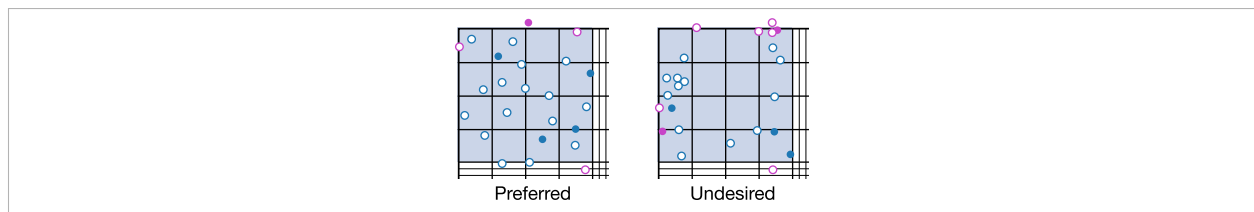
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Resources

Cell counting with a Neubauer cell chamber



In Step 5: The design of the nine large $100 \mu\text{m}^2$ grids varies between hemocytometer models.



In Step 5: Particles touching the top or right borders of a grid are counted; those touching the bottom or left are excluded. Uniform cell distribution is essential.

Change log

2000-01-01	Unknown	Initial protocol; Jue Chen lab.
2015-08-07	Martin Wilson	Haemocytometer types; online resource (Agar Scientific Ltd.).
2020-05-27	Michael Haugbro	Initial protocol; Tom Muir lab.
2023-03-14	Benjamin C. Buchmuller	Adaptation as SOP.

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